

Homework 09.14.26 Question 2

2026-09-13

Causal inference (Exercise 1.6 of Regression and Other Stories). Find a real-world example of interest with a treatment group, control group, a pre-treatment predictor, and a post-treatment predictor. Make a graph like Figure 1.8 using the data from this example.

I chose to use LaLonde's National Supported Work study — it is available through the MatchIt package in R. The National Supported Work (NSW) experiment was designed to test whether participation in a job-training program improved participants' later labor-market outcomes, especially earnings. Individuals were assigned either to receive the training program (the treatment group) or to not (the control group). Their earnings were compared before (1975) and after (1978) the program.

To mimic Figure 1.8, I generated predicted outcomes for treated and control observations across a range of 1975 earnings, holding other covariates constant, and used the difference between the two predicted lines as the estimated treatment effect.

```
library(ggplot2)
library(MatchIt)

data("lalonde", package = "MatchIt")

# Fit a regression predicting 1978 earnings using treatment status, prior earnings,
# and other pre-treatment background variables like age, education level, race,
# and marital status.
model <- lm(
  re78 ~ treat + re75 + re74 + age + educ + race + married + nodegree,
  data = lalonde
)

# Create a range of 1975 earnings values to plot along the x-axis. At each value,
# calculate a predicted 1978 earnings value for both the control group (treat = 0)
# and treatment group (treat = 1). All other background variables are held constant.
prediction_data <- expand_grid(
  re75 = seq(
    min(lalonde$re75),
    max(lalonde$re75),
    length.out = 100
  ),
  treat = c(0, 1),
  re74 = mean(lalonde$re74),
  age = mean(lalonde$age),
  educ = mean(lalonde$educ),
  race = "black",
  married = 0,
  nodegree = 1
)
```

```

# Use the fitted regression model to predict 1978 earnings across a range of
# 1975 earnings values.
prediction_data$predicted_re78 <- predict(
  model,
  newdata = prediction_data
)

# Create two otherwise identical observations at re75 = $17,000, one for the
# treatment group and one for the control group. These will be used to create
# the arrow showing the treatment effect.
x_arrow <- 17000

arrow_data <- data.frame(
  re75 = c(x_arrow, x_arrow),
  treat = c(0, 1),
  re74 = c(mean(lalonde$re74), mean(lalonde$re74)),
  age = c(mean(lalonde$age), mean(lalonde$age)),
  educ = c(mean(lalonde$educ), mean(lalonde$educ)),
  race = c("black", "black"),
  married = c(0, 0),
  nodegree = c(1, 1)
)

# Predict 1978 earnings for the control and treated observations defined above.
# The resulting values are scaled to thousands of dollars below to match the graph.
arrow_predictions <- predict(
  model,
  newdata = arrow_data
)

control_y <- arrow_predictions[1] / 1000
treatment_y <- arrow_predictions[2] / 1000

# Calculate the estimated treatment effect in dollars.
treatment_effect <- arrow_predictions[2] - arrow_predictions[1]

# All model predictions and the treatment effect have now been calculated.
# All code below visualizes the observed data and these model results.

ggplot(
  data = lalonde,
  mapping = aes(
    x = re75 / 1000,
    y = re78 / 1000
  )
) +

# Plot the observed data.
geom_point(
  aes(shape = factor(treat)),
  size = 2
) +

```

```
# Draw the two fitted regression lines using the model's predicted values.
```

```
geom_line(  
  data = prediction_data,  
  aes(  
    x = re75 / 1000,  
    y = predicted_re78 / 1000,  
    group = factor(treat)  
  ),  
  color = "black",  
  linewidth = 0.7  
) +
```

```
# Draw an arrow showing the vertical distance between the fitted treatment  
# and control lines, visually depicting the estimated treatment effect.
```

```
annotate(  
  "segment",  
  x = x_arrow / 1000,  
  xend = x_arrow / 1000,  
  y = control_y,  
  yend = treatment_y,  
  color = "red",  
  linewidth = 0.7,  
  arrow = grid::arrow(  
    ends = "both",  
    type = "closed",  
    angle = 25,  
    length = grid::unit(0.08, "cm")  
  )  
) +
```

```
# Label the treatment effect.
```

```
annotate(  
  "text",  
  x = 19,  
  y = max(control_y, treatment_y) + 2,  
  label = paste0(  
    "Estimated treatment\\neffect = $",  
    round(treatment_effect, 2)  
  ),  
  size = 3  
) +
```

```
# Label the two fitted lines directly, similar to Figure 1.8.
```

```
annotate(  
  "text",  
  x = 14,  
  y = 7.4,  
  label = "Controls",  
  size = 3.5  
) +
```

```
annotate(  
  "text",
```

```

    x = 14,
    y = 10.5,
    label = "Treated",
    size = 3.5
  ) +

  # Set point shapes: treatment is open, unfilled circles, and control is solid,
  # filled circles. Then, create a legend.
  scale_shape_manual(
    values = c(
      "0" = 16,
      "1" = 1
    ),
    labels = c(
      "0" = "Control",
      "1" = "Treatment"
    ),
    name = NULL
  ) +

  # Set both axes to begin at zero (remove the padding that is included in ggplot by
  # default).
  scale_x_continuous(
    limits = c(0, max(lalonde$re75 / 1000)),
    expand = c(0, 0)
  ) +

  scale_y_continuous(
    limits = c(0, max(lalonde$re78 / 1000)),
    expand = c(0, 0)
  ) +

  labs(
    title = "Job Training and Earnings",
    x = "Pre-treatment earnings in 1975 ($1,000s)",
    y = "Post-treatment earnings in 1978 ($1,000s)",
    caption = "Source: LaLonde (1986), National Supported Work (NSW) Dataset"
  ) +

  # Zoom in so the difference between the fitted lines is more visually distinct.
  coord_cartesian(
    ylim = c(0, 20)
  ) +

  theme_classic() +

  # Center the title and place the legend below the graph.
  theme(
    plot.title = element_text(hjust = 0.5),
    plot.title.position = "plot",
    legend.position = "bottom"
  )

```



Source: LaLonde (1986), National Supported Work (NSW) Dataset